

SOLUTIONS ANSWER KEY

Exercise: 1

IAT Based Problems

1. In the case of zinc amalgam, which is an alloy of zinc with mercury, both zinc and mercury are typically in the liquid state at room temperature. Therefore, the physical state of the solute (zinc) is liquid, and the physical state of the solvent (mercury) is also liquid.

Ans. B.

2. $\text{No. of moles} = \frac{\text{Given mass of substance}}{\text{Molar mass of substance}}$

In this case, you're given the volume (2L) and the molarity (0.5 M) of sodium hydroxide (NaOH). The molarity (M) is defined as moles of solute per litre of solution.

$$\text{Molarity} = \frac{\text{number of moles}}{\text{volume (in litres)}}$$

Rearranging the formula to find the no. of moles:

$$\text{No. of moles} = \text{Molarity} \times \text{Volume (in litres)}$$

$$\text{No. of moles} = 0.5 \text{ mol/L} \times 2\text{L} = 1 \text{ mol}$$

Ans. C.

3. Molal concentration (molality) is defined as the number of moles of solute per kilogram of solvent. The formula for molality (m) is given by:

$$\text{Molality (m)} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent (in kg)}}$$

Molality is a useful concentration unit, especially when dealing with temperature changes, because it is not temperature-dependent, unlike molarity.

Ans. B.

4. To convert molality into Percentage by volume unit of concentration requires the density of the solution.

Ans. B.

5. Given $M_1 = 3 \text{ M}$, $V_1 = 25 \text{ ml}$, $M_2 = 4 \text{ M}$, $V_2 = 75 \text{ ml}$

$$M_3V_3 = M_1V_1 + M_2V_2$$

$$M_3(V_1+V_2) = M_1V_1 + M_2V_2$$

$$M_3(25+75) = 3 \times 25 + 4 \times 75$$

$$M_3 = \frac{375}{100} = 3.75 \text{ M}$$

Ans. C.

6. $M = \frac{W_2 \times 100}{Mw_2 \times V_{sol}}$ (1 decilitre = 100 ml)

$$M = \frac{40 \times 1000}{40 \times 100} = 1$$

$$m = \frac{W_2 \times 1000}{Mw_2 \times (\text{weight of soln} - \text{weight of solute})}$$

W_1 = weight of solvent

= weight of solution - weight of solute

$$= V_{sol} \times d_{sol} - W_2$$

$$= 100 \times 1.038 - 4 = 99.8 \text{ g}$$

$$\therefore m = \frac{4 \times 1000}{40 \times 99.8} = 1.002$$

Ans. B.

7. $M = \frac{240}{60 \times 10} = 0.4$

Ans. C.

8. The vapour pressure of a liquid is a function of temperature. As the temperature decreases, the kinetic energy of the molecules in the liquid decreases, and fewer molecules have the energy required to overcome the attractive forces and enter the vapour phase. Therefore, lowering the temperature leads to a decrease in vapour pressure. A, B, and C are not directly related to the temperature effect on vapour pressure:

Ans. D.

9. Raoult's law is satisfied when the forces of attraction between like molecules (of the same component) are identical to the forces between unlike molecules (of different components). According to Raoult's law, the partial vapour pressure of each component in the solution is directly proportional to its mole fraction in the mixture and the vapour pressure of the pure component.

If the forces of attraction between like molecules are identical to those between unlike molecules, the behaviour of the components in the solution follows Raoult's law. This implies that the components interact with each other in a similar manner, allowing for an ideal solution.

Ans. C.

10. The substance with the lowest vapour pressure is mercury (B). Mercury, being a heavy metal with weak intermolecular forces, has a higher vapour pressure compared to water, ethanol, and acetone.

Ans. B.

11. $\Delta G = \Delta H - T\Delta S$

For ideal, $\Delta H = 0$

$$\Delta G = 0 - T\Delta S$$

ΔG is determined by ΔS

$$\Delta G = -T\Delta S \quad (\Delta S = +ve)$$

$$\Delta G = -ve$$

Ans. C.

12. An ionic compound that attracts atmospheric water so strongly that a hydrate is formed is referred to as hygroscopic. This property is common in certain salts and substances that have a strong affinity for water molecules, causing them to absorb moisture from the surrounding environment.

Ans. B.

13. A solution of sulphuric acid in water exhibits negative deviations from Raoult's law. Sulphuric acid and water form mixtures in which particles break away to form the vapour with much more difficulty than in either of the pure liquids.

Ans. B.

14. Osmosis results from an increase in entropy as pure solvent passes through the membrane and mixes with the solution. And Desalination of sea water is done by reverse osmosis.

Ans. C.

15. Depression of the freezing point is associated with the disorder in a solution, reflected by an increase in entropy. The addition of solute disrupts the regular arrangement of solvent molecules, leading to higher entropy in the solution compared to the pure solvent.

Ans. A.

NEST Based Problems

1. The volume change on mixing is zero: This statement is generally true for an ideal solution. In ideal solutions, the molecules of the components are assumed to have similar sizes and interact similarly with each other, leading to no significant volume change upon mixing.

Ans. A.

2. $M_1V_1 = M_2V_2$

$$100 \times 0.1 = x \times 0.2$$

$$x = \frac{100 \times 0.1}{0.2} = 50 \text{ mL}$$

To make the required solution, we have to reduce the volume of the solution to 50 mL.

Hence, this can be done when 50 mL of water or solution is evaporated. Therefore, use the formula of molarity to evaporate it.

From equation (1), we get –

$$\text{Molarity} = \frac{n}{V}$$

$$\Rightarrow \text{Molarity} = \frac{n}{0.1}$$

$$\Rightarrow 0.2M = \frac{n}{0.1}$$

$$\Rightarrow n = 0.02M$$

So, the number of moles should be increased from 0.01 to 0.02 by adding the 0.01 moles of KCl.

Hence, to make the solution of KCl 0.2 M, we have to evaporate the 50 mL water and 50 mL solution and add 0.01 mol KCl.

Ans. C.

3. $\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$

$$0.05M \quad 0.05M \quad 0.05M$$

$$\text{Total } C_1 = 0.05 + 0.05 = 0.1\text{M (NaCl)}$$

$$C_2 = 0.2\text{M (glucose)}$$

$$\pi = (C_2 - C_1)RT$$

$$\pi = (0.2 - 0.1) \times 0.083 \times 300$$

$$= 2.49 \text{ bar}$$

$$= 24.9 \times 10^{-1} \text{ bar}$$

Ans. A.

4. The elevation in boiling point is:

$$\Delta T_b = K_b \times m$$

$$m = \text{molality} = (n_2 / w_1) \times 1000$$

$$[n_2 = \text{Number of moles of solute, } w_1 = \text{Weight of solvent in gram}]$$

$$2 = 0.76 \times (n_2 / 100) \times 1000$$

$$n_2 = 5 / 19$$

Also, from Raoult's law of lowering of vapour pressure:

$$\Delta p / p^0 = x_2 = n_2 / (n_1 + n_2) \approx n_2 / n_1 \quad [\text{since } n_1 \gg n_2]$$

$$\Delta p = 760 \times (5 / 19) \times (18 / 100)$$

$$\Delta p = 36 \text{ mm of Hg}$$

$$p = 760 - 36 = 724 \text{ mm of Hg}$$

Ans: B

5.

Boiling point elevation constant: $K_b = 0.5 \text{ K kg mol}^{-1}$

Freezing point depression constant: $K_f = 1.8 \text{ K kg mol}^{-1}$

Change in boiling point: $\Delta T_b = 100.15^\circ\text{C} - 100^\circ\text{C} = 0.15^\circ\text{C}$

Change in freezing point: $\Delta T_f = 0^\circ\text{C} - (-0.8^\circ\text{C}) = 0.8^\circ\text{C}$

Amount of NaCl added: 0.2 mol

Temperature: 298 K

Step 1: Calculate the molality using boiling point elevation

The formula gives the boiling point elevation:

$$\Delta T_b = K_b \times m$$

Where m is the molality.

Substituting the values:

$$0.15 = 0.5 \times m$$

$$m = 0.15 / 0.5 = 0.3 \text{ mol/kg}$$

Step 2: Calculate molality using freezing point depression

The freezing point the formula gives depression:

$$\Delta T_f = K_f \times m$$

Substituting the values:

$$0.8 = 1.8 \times m$$

$$m = 0.8 / 1.8 \approx 0.444 \text{ mol/kg}$$

Step 3: Solubility product calculation

For lead chloride (PbCl_2), it dissociates as:



The formula gives the solubility product K_{sp} :

$$K_{sp} = [\text{Pb}^{2+}] \times [\text{Cl}^-]^2$$

From the molality values, we estimate the concentrations of Pb^{2+} and Cl^- , and using the dissociation, we compute K_{sp} . After the computation, the nearest value of K_{sp} is:

$$K_{sp} = 13 \times 10^{-6}$$

Ans: C

6.

Vapour pressure of pure methanol (A): When the mole fraction of methanol, $x=1$ (pure methanol), the equation becomes:

$$p = 119(1) + 135 = 119 + 135 = 254 \text{ torr}$$

Vapour pressure of pure ethanol (B): When the mole fraction of methanol, $x=0$ (pure ethanol), the equation becomes:

$$p = 119(0) + 135 = 135 \text{ torr}$$

vapour pressure of an equimolar mixture of each (C): In an equimolar mixture of methanol and ethanol, $x=0.5$. Substituting this into the equation:

$$p = 119(0.5) + 135 = 59.5 + 135 = 194.5 \text{ torr}$$

Mixture completely immiscible (D): Methanol and ethanol are miscible, so **Option D** is **incorrect**.

Ans: B

7.

Explanation: Raoult's Law is used to calculate the total vapour pressure of a mixture of two volatile liquids. The partial pressure of each component is proportional to its mole fraction in the liquid phase and its vapour pressure when pure. In this case, the total vapour pressure is calculated by adding the contributions from both A and B. The mole fraction of component B in the vapour phase is found using the ratio of its partial pressure to the total vapour pressure.

The total vapour pressure (P_{Total}) is calculated using Raoult's Law:

$$P_T = P_A^0 x_A + P_B^0 x_B$$

Given that the vapour pressure of liquid A is 50 torr and that of liquid B is 100 torr, we are also given the mole fractions of A and B in the liquid phase as $x_A = 0.3$ and $x_B = 0.7$, respectively.

Therefore, we calculate the total vapour pressure as:

$$P_T = 50 \times 0.3 + 100 \times 0.7 = 85 \text{ torr}$$

Next, to calculate the mole fraction of component B in the vapour phase, we use the equation:

$$P_2 = y_B \times P_T$$

Given that the partial pressure of component B is 70 torr, we solve for y_B :

$$70 = yB \times 85$$

$$yB = 70 / 85 = 14 / 17$$

Thus, the mole fraction of liquid B in the vapour phase is 14/17. Therefore, the values of x and y are as follows:

$$x = 14; y = 17$$

Ans: C

8. Density of solution = 1.110 g mL^{-1}

Concentration = 3.0 molal

This implies 1000 g of water has 3 moles of NaOH

3 moles of NaOH = $3 \times 40 = 120 \text{ g}$ of NaOH

Mass of solution = $1000 + 120 = 1120 \text{ g}$

The volume of solution = $\frac{\text{mass}}{\text{density}} = \frac{1120}{1.11} = 1009 \text{ mL}$

Molarity = $\frac{\text{no. of moles}}{\text{volume in L}} = \frac{3}{\frac{1009}{1000}} = 2.97 \text{ M}$

Ans. D.

9. Mole fraction is the number of moles of part over total moles or

$$\text{mole fraction} = \frac{\text{no. of moles of a solute}}{\text{total no. of moles in solution}}$$

If, for eg., we take that no. of moles of solute as n_1 and that of solvent is n_2

Mole fraction of solute = $\frac{n_1}{n_1 + n_2}$

Mole fraction of solvent = $\frac{n_2}{n_1 + n_2}$

The sum of mole fractions = $\frac{n_1}{n_1 + n_2} + \frac{n_2}{n_1 + n_2} = 1$

Ans. C

10. 10 N $\rightarrow$ decanormal

$\frac{1}{10}$ N $\rightarrow$ decinormal

Ans. D.

11. 10% $\left(\frac{w}{v}\right)$ acetic acid means 10 g acetic acid in 100 ml water.

Thus, 100 g of acetic acid would be present in 1000 ml of water.

60 g = 1 mole of acetic acid

$$\therefore 100 \text{ g} = \frac{100}{60} = 1.67 \text{ moles of } \text{CH}_3\text{CO}_2\text{H}.$$

Thus, 1.67 moles are present in 1L of water, i.e., the solution is 1.67 M.

In case of $\text{CH}_3\text{CO}_2\text{H}$, 1N solution = 1M solution

Thus, the normality of 10% acetic acid = 1.67

Ans. C

12. A solution of CaCl_2 is 0.5 mol/L

0.500 L of solution will contain = $0.500 \times 0.5 = 0.25$ moles of CaCl_2

1 mole of $\text{CaCl}_2 = 2$ moles of Cl ions

0.25 moles of $\text{CaCl}_2 = 2 \times 0.25 = 0.50$ moles of Cl ions

Ans. B.

13. Mole fraction = $\frac{\text{moles of one component}}{\text{total moles of all component}}$

$\therefore$ Sum of mole fractions of all components in solution is always

$$\text{one, i.e., } \chi_A + \chi_B + \chi_C = \frac{n_A}{n_A+n_B+n_C} + \frac{n_B}{n_A+n_B+n_C} + \frac{n_C}{n_A+n_B+n_C} = 1$$

Ans. C.

14. Equivalent wt of dibasic acid = $\frac{\text{mol wt}}{2} = \frac{200}{2} = 100$

Strength = 0.1 N, m = ?, V = 100 mL

$$\text{From, } m = \frac{EVN}{1000} = \frac{100 \times 100 \times 0.1}{1000} = 1 \text{ g}$$

Ans. A

15. Colligative properties specifically refer to properties of a solution that depend on the concentration of solute particles, such as vapour pressure lowering, boiling point elevation, freezing point depression, and osmotic pressure. These properties are related to the number of solute particles rather than their nature.

Ans. D

16. According to lowering of vapour pressure

$$\frac{P^0 - P}{P^0} = \chi_2$$

Where P^0 = vapour pressure of pure solvent

P = vapour pressure of solute

χ_2 = mole fraction of the solute

$$\frac{50 - 45}{50} = \chi_2$$
$$\chi_2 = 0.100$$

Ans. C

17. The elevation of boiling point method is used for the determination of the molecular weight of a non-volatile and soluble solute.

Ans. A.

18. The highest temperature at which the vapour pressure of a liquid can be measured is the boiling point because at the boiling point, the vapour pressure is equal to atmospheric pressure. With any incremental increase in that temperature, the vapour pressure becomes sufficient to overcome atmospheric pressure and lift the liquid to form vapour bubbles inside the bulk of the substance.

Ans. D.

19. Due to the solute presence in the solution, the solution's boiling point is greater than the pure solvent's boiling point. This is called elevation in boiling point, and it can be represented as:

$$\Delta T_b = K_b \times \text{molality}$$
$$K_b = \frac{\Delta T_b}{\text{molality}}$$

Ans. B.

21. The elevation of a boiling point on adding one mole solute to 1000 g solvent (one molal) is called the molal boiling point elevation constant. It is denoted by K_b . Its unit is K kg mol^{-1} , e.g., for water, $K_b = 0.52 \text{ K kg mol}^{-1}$.

Ans. B.

22. Here are the freezing points of the substances in question:

- **Water:** 0°C (273 K)
- **Ethanol:** -114°C (159 K)
- **Nitrogen:** -210°C (63 K)

- **Carbon dioxide:** -78°C (195 K)

Among these, **nitrogen** has the lowest freezing point at -210°C , making it the correct answer.

Ans. C.

23. Mass % of $\text{H}_2\text{SO}_4 = 29\%$

100 g solution contains 29 g H_2SO_4

$$\text{Molarity} = \frac{\text{Moles of } \text{H}_2\text{SO}_4}{\text{Volume of solution (ml)}} \times 1000$$

$$M = 3.60$$

$$\frac{\frac{29}{98}}{\frac{100}{d}} \times 1000 = 3.6$$

$$d = 1.22 \text{ g/ml}$$

Ans. D

$$24. N_{\text{HCl}} = \frac{W_B \times 1000}{E_B \times V} = \frac{0.04 \times 1000}{36.5 \times 1} = 1.095$$

$$N_{\text{NaOH}} = N_{\text{HCl}}$$

$$\therefore 1.095 = \frac{W_B \times 1000}{40 \times 1}$$

$$W_B = 0.0438 \text{ g/mL}$$

Ans. D

25. The number of moles of nitric acid can be obtained by multiplying the molarity with volume

$$M \times V = \text{moles of } \text{HNO}_3 = \frac{250 \times 2}{1000} = 0.5$$

Ans. A.

26. $\Delta T_f = K_f \times \text{molality}$

$$\Rightarrow 0 - (-10) = 1.86 \times \frac{W}{5.5}$$

$$\Rightarrow 5.5 \times 10 = \frac{1.86}{100} \times \frac{W}{62}$$

$$\Rightarrow 55 = \frac{3}{100} \times W$$

$$\Rightarrow W = \frac{55 \times 100}{3} = 18.33 = 1833 \text{ g} = 1.833 \text{ kg}$$

Ans. B.

27. $W_b = 15 \text{ g}$

$M_b = ?$

The freezing point of water = 273 K

Freezing point of solution = $-0.34 + 273 \text{ K}$

Depression in freezing point = $273 - (-0.34 + 273) = 0.34 \text{ K}$

K_f for water = $1.86 \text{ K kg mol}^{-1}$

$$\Delta T_f = K_f \times m = K_f \frac{W_b \times 1000}{M_b \times W_a}$$

$$M_b = \frac{K_f \times w_b \times 1000}{w_a \times \Delta T_f}$$

$$M_b = \frac{1.86 \times 15 \times 1000}{450 \times 0.34} = 182.35 \text{ g/mol}$$

Ans. C.

28. The formula for calculating the molality of the solution if an elevated boiling point is given is:

$$\Delta T = K_b m$$

K_b water = $0.51 \text{ } ^\circ\text{C kg/mol}$ (standard value)

Given,

$$\Delta T = 100.42^\circ\text{C} - 100^\circ\text{C} = 0.42^\circ\text{C}$$

$$m = \Delta T / K_b = 0.42 / 0.51 = 0.8$$

$$\text{Molality} = \frac{\text{Number of moles}}{\text{mass of solvent in gram}} \times 1000$$

$$\text{Molality} = \frac{\text{given mass of glycerol}}{\text{molecular mass of glycerol}} \times \frac{1000}{\text{mass of water in g}}$$

$$0.8 = \frac{\text{given mass of glycerol}}{92} \times \frac{1000}{500}$$

$$\text{Given mass of glycerol} = 0.8 \times 92 \times \frac{500}{1000} = 37.7 \text{ g}$$

The mass of glycerol used to make the above solution is 37.7 g.

Ans. D.

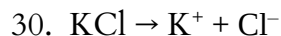
29. $\Delta T = K_b m$

$$\Rightarrow 80.31 - 80.10 = 2.53 \times \frac{\frac{1.25}{M} \times 100}{99}$$

$$\Rightarrow 0.21 = 2.56 \times \frac{1250}{99M}$$

$$\Rightarrow M = \frac{2.53 \times 1250}{99 \times 0.21} = 152 \text{ g/mol}$$

Ans. C



$$n = 2$$

$$i = 1 - \alpha + n\alpha$$

$$i = 1 + \alpha$$

$$\Delta T_f = iK_f m$$

$$= (1 + 0.92) \times 1.86 \times \frac{0.5 \times 1000}{74.5 \times 100}$$

$$\Delta T_f = 0.24$$

$$\Delta T_f = T_f^0 - T_f$$

$$T'_f = -0.24^\circ\text{C}$$

Ans. D

31. $i = 2$

$$W_A = 1 \text{ kg} = 1000 \text{ g}$$

$$K_f = 1.86 \text{ K kg mol}^{-1}$$

$$H_B = 74.5 \text{ g/mol}$$

Substituting those values in expansion

$$W_B = \frac{\Delta T_f \times H_B \times W_A}{iK_f \times 1000} = \frac{2 \times 74.5 \times 1000}{2 \times 1.86 \times 1000} = 40.05 \text{ gm}$$

Ans. C

32. In the case of the AB_2 compound

$$M_B = \frac{K_f W_B \times 1000}{W_A \Delta T_f}$$

$$M_B = \frac{5.1 \times 1 \times 1000}{20 \times 2.3} = 110.87 \text{ g/mol}$$

In the case of compound AB_4

$$M_B = \frac{5.1 \times 1 \times 1000}{20 \times 1.3} = 196.15 \text{ g/mol}$$

Let a g/mol and b g/mol be the atomic masses of A and B, respectively.

$$M_{\text{AB}_2} = a + 2b = 110.87 \quad \text{---(i)}$$

$$M_{\text{AB}_4} = a + 4b = 196.15 \quad \text{---(ii)}$$

Subtracting equation (ii) from equation (i), we have

$$-2b = -85.28$$

Atomic mass of B is $b = 42.64 \text{ g/mol}$

Substituting the value of b in equation (i), we get

$$a + 2 \times 42.64 = 110.87$$

The atomic mass of A is $a = 25.59 \text{ g/mol}$

Ans. D.

33. Given:

Vapour pressure of pure liquid A, $P_A^0 = 450 \text{ mm of Hg}$

Vapour pressure of pure liquid B, $P_B^0 = 700 \text{ mm of Hg}$

Total vapour pressure, $P_{total} = 600 \text{ mm of Hg}$

Use the formula of Raoult's Law

$$600 = (450 - 700)\chi_A + 700$$

$$250\chi_A = 100$$

$$\chi_A = \frac{100}{250} = 0.4$$

Use formula

$$\chi_B = 1 - \chi_A$$

Substitute the values we get, $\chi_B = 1 - 0.4 = 0.6$

Use formula $P_A = P_A^0 \times \chi_A = 450 \times 0.4 = 180 \text{ mm of Hg}$

$$P_B = P_B^0 \times \chi_B = 700 \times 0.6 = 420 \text{ mm of Hg}$$

Now, in the vapour phase:

$$\text{Mole fraction of liquid A} = \frac{180}{180+420} = 0.30$$

$$\text{Mole fraction of liquid B, } Y_B = 1 - Y_A = 1 - 0.30 = 0.70$$

Ans. A

34. 5% by mass of cane sugar means 100 g of solution has 5 g of cane sugar. The molecular weight of cane sugar is 342 g/mol.

$$\text{Number of moles of cane sugar} = \frac{5}{342} = 0.0146 \text{ mol}$$

$$\text{Mass of water} = 100 \text{ g} - 5 \text{ g} = 95 \text{ g} = 0.095 \text{ kg}$$

$$(\because 1 \text{ kg} = 1000 \text{ g})$$

$$\text{Molality } m = \frac{0.0146}{0.095} = 0.154 \text{ mol/kg}$$

The depression at the freezing point

$$\Delta T_f = 273.15 \text{ K} - 271 \text{ K} = 2.15 \text{ K}$$

$$\Delta T_f = K_f m$$

$$K_f = \frac{\Delta T_f}{m} = \frac{2.15}{0.154} = 13.97 \text{ K kg/mol}$$

5% by mass of glucose means 100 g of solution has 5 g of glucose. The molecular weight of glucose is 180 g/mol.

$$\text{Number of moles of glucose} = \frac{5}{180} = 0.0277 \text{ mol}$$

$$\text{Mass of water} = 100 \text{ g} - 5 \text{ g} = 95 \text{ g} = 0.095 \text{ kg}$$

$$(\because 1 \text{ kg} = 1000 \text{ g})$$

$$\text{Molality } m = \frac{0.0277}{0.095} = 0.292 \text{ mol/kg}$$

The depression in the freezing point

$$\Delta T_f = K_f m$$

$$\Delta T_f = \frac{0.292}{13.97} \times 0.292 = 4.09 \text{ K}$$

The freezing point of glucose solution = $273.15 \text{ K} - 4.09 \text{ K} = 269.07 \text{ K}$

Ans. A.

35. Mass of sucrose (W_B) = 4g

Mass of water (W_A) = $100 - 4 = 96 \text{ g}$

$M_B = 342$

$$\Delta T_f = 273.15 - 271.5 = 2 \text{ K}$$

$$K_f = \frac{\Delta T_f \times M_B \times W_A}{W_B \times 1000}$$

$$K_f = \frac{2 \times 342 \times 96}{4 \times 1000} = 16.416$$

Now, for the solution of glucose

Mass of glucose (W_B) = 5 g

Mass of water (W_A) = $100 - 5 = 95$

$M_B = 180 \text{ g/mol}$

$$\Delta T_f = \frac{K_f \times W_B \times 1000}{M_B \times W_A} = \frac{16.416 \times 5 \times 1000}{180 \times 95} = 4.8 \text{ K}$$

$$\Delta T_f = T_f^0 - T_f = 4.8 = 273.15 - T_f$$

$$T_f = 273.15 - 4.8 = 268.35 \text{ K}$$

Ans. C

Exercise: 2

IAT PYQs

1.

The given system involves partial pressures, where:

$$P_t = X_1 P_1^\circ + X_2 P_2^\circ$$

Step 1: The total pressure equation is given by:

$$500 = \frac{1}{3} P_1^\circ + \frac{2}{3} P_2^\circ$$

Step 2: Another equation is provided:

$$P_1^\circ + 2P_2^\circ = 1500$$

Rearranging this equation:

$$2P_2^\circ = 1500 - P_1^\circ$$

$$P_2^\circ = (1500 - P_1^\circ) / 2$$

Step 3: Substituting P_2° into the first equation:

$$500 = \frac{1}{3} P_1^\circ + \frac{2}{3} [(1500 - P_1^\circ) / 2]$$

Multiplying the entire equation by 3 to eliminate the fractions:

$$1500 = P_1^\circ + 2P_2^\circ$$

This confirms the earlier equation:

$$P_1^\circ + 2P_2^\circ = 1500$$

Step 4: Solving for P_1°

Substitute into the second equation:

$$2625 = 3P_1^\circ + 2P_2^\circ$$

Substitute $P_2^\circ = (1500 - P_1^\circ) / 2$ into this equation:

$$2625 = 3P_1^\circ + (1500 - P_1^\circ)$$

Simplifying:

$$2625 = 2P_1^\circ + 1500$$

$$2P_1^\circ = 1125$$

$$P_1^\circ = 562.5$$

Step 5: Solving for P_2°

Now substitute $P_1^\circ = 563$ into the equation $P_1^\circ + 2P_2^\circ = 1500$:

$$563 + 2P_2^\circ = 1500$$

$$2P_2^\circ = 937$$

$$P_2^\circ = 468.5$$

The values obtained are $P_1^\circ \approx 563$ and $P_2^\circ \approx 469$.

Ans. A.

2.

We are given that the total volume of the NaCl solution is 1000 mL, and the density of the solution is 1.25 g/mL. The mass of the solution can be calculated as:

$$\text{Mass of solution} = \text{density} \times \text{volume} = 1.25 \text{ g/mL} \times 1000 \text{ mL} = 1250 \text{ g}$$

$$\text{Molarity} = \text{moles of solute} / \text{volume of solution (L)}$$

$$3 \text{ M} = \text{moles of NaCl} / 1 \text{ L}$$

$$\text{Moles of NaCl} = 3 \text{ moles}$$

$$\text{Mass of NaCl} = \text{moles} \times \text{molar mass} = 3 \text{ mol} \times 58.5 \text{ g/mol} = 175.5 \text{ g}$$

Thus, the mass of water is:

$$\text{Mass of water} = \text{total mass} - \text{mass of NaCl} = 1250 \text{ g} - 175.5 \text{ g} = 1074.5 \text{ g}$$

Since the density of water is 1 g/mL, the volume of water required is:

$$\text{Volume of water} = \text{mass} / \text{density} = 1074.5 \text{ g} / 1 \text{ g/mL} = 1074.5 \text{ mL}$$

Therefore, the correct answer is 1074.5 mL

Ans. A.

3. For the first solution X (hexane and benzene), both hexane and benzene are non-polar and have similar intermolecular forces. As a result, they follow Raoult's law and exhibit ideal behaviour.

For the second solution, Y (water and HCl), strong interactions such as hydrogen bonding and ion-dipole forces between water and HCl lead to a negative deviation from Raoult's law

Thus, the correct answer is A. No deviation (ideal behaviour) and negative deviation, respectively.

Ans. A.

4.

Osmotic pressure (Π) is given by the formula:

$$\Pi = nRT / V$$

Where:

- n is the number of moles of solute
- R is the gas constant
- T is the absolute temperature
- V is the volume of the solution

From this equation, we see that osmotic pressure (Π) is inversely proportional to the volume (V). As the volume increases, the osmotic pressure decreases, following a hyperbolic decay. This means that the correct plot should show a hyperbolic curve.

Option A shows a hyperbolic decay, which correctly represents the inverse relationship between osmotic pressure and volume.

Ans. A.

NEST PYQs

1.

The formula for freezing point depression is given by:

$$\Delta T_f = K_f \times m$$

Where:

ΔT_f is the freezing point depression,

K_f is the cryoscopic constant for water ($K_f = 1.86^\circ\text{C kg/mol}$),

m is the molality of the solution.

The molar mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is approximately 180 g/mol. The number of moles of glucose is given by:

$$\begin{aligned}\text{Moles of glucose} &= \text{Mass of glucose} / \text{Molar mass of glucose} \\ &= 18 \text{ g} / 180 \text{ g/mol} = 0.1 \text{ mol}\end{aligned}$$

The molality of the glucose solution is given by:

$$\begin{aligned}m &= \text{Moles of solute/kg of solvent} \\ &= 0.1 \text{ mol} / X \text{ kg of water}\end{aligned}$$

The freezing point depression for the glucose solution is:

$$\Delta T_x = 1.86 \times (0.1 / X) = 0.186 / X$$

NaCl dissociates into Na^+ and Cl^- ions, so the van't Hoff factor (i) for NaCl is 2. Let the mass of NaCl required be m_{NaCl} . The molar mass of NaCl is approximately 58.5 g/mol. The number of moles of NaCl is given by:

$$\text{Moles of NaCl} = m_{\text{NaCl}} / 58.5$$

The molality of the NaCl solution is:

$$m = (m_{\text{NaCl}} / 58.5) / X$$

The freezing point depression for NaCl is:

$$\Delta T_x = 2 \times 1.86 \times (m_{\text{NaCl}} / 58.5X) = (3.72 \times m_{\text{NaCl}}) / 58.5X$$

Since the freezing point depression for both glucose and NaCl solutions is the same, we equate the two expressions:

$$0.186 / X = (3.72 \times m_{\text{NaCl}}) / 58.5X$$

Solving for m_{NaCl} :

$$m_{\text{NaCl}} = (0.186 \times 58.5) / 3.72 \approx 2.93 \text{ g}$$

Ans: D.

The van't Hoff factor (i) is the number of particles (ions or molecules) a solute produces when it dissociates in solution. The boiling point elevation is directly proportional to this factor.

Sucrose ($C_{12}H_{22}O_{11}$):

Sucrose is a non-electrolyte and does not dissociate in water. Therefore, the van't Hoff factor (i) for sucrose is 1.

KBr (Potassium bromide):

KBr is an ionic compound that dissociates into 2 ions: K^+ and Br^- . Therefore, the van't Hoff factor (i) for KBr is 2.

$K_4[Fe(CN)_6]$ (Potassium hexacyanoferrate):

$K_4[Fe(CN)_6]$ dissociates into 5 ions: 4 K^+ ions and 1 $[Fe(CN)_6]^{4-}$ ion. Therefore, the van't Hoff factor (i) for this compound is 5.

K_2CO_3 (Potassium carbonate):

K_2CO_3 dissociates into 3 ions: 2 K^+ ions and 1 CO_3^{2-} ion. Therefore, the van't Hoff factor (i) for K_2CO_3 is 3.

Boiling Point Elevation Formula:

The boiling point elevation (ΔT_b) is given by the formula:

$$\Delta T_b = K_b \times m \times i$$

Where:

- ΔT_b is the elevation in boiling point
- K_b is the ebullioscopic constant (same for all solutions)
- m is the molality (same for all)
- i is the van't Hoff factor.

Order of Boiling Point Elevation Based on i :

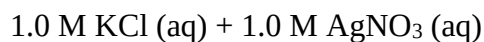
Based on the van't Hoff factor (i) values:

- Sucrose: $i = 1$
- KBr: $i = 2$
- $K_4[Fe(CN)_6]$: $i = 5$
- K_2CO_3 : $i = 3$

The correct order of boiling point elevation is:
 $K_4[Fe(CN)_6] > K_2CO_3 > KBr > \text{Sucrose}$.

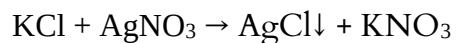
Ans. B.

3.



$$\Delta T_f = ?$$

$$K_{f_{H_2O}} = 1.86 \text{ K kg mol}^{-1}$$



$$i = \frac{v}{v+v} = \frac{1}{2}$$

$$\text{Concentration} = \frac{n}{V}$$

$$m_{sol} = m_{KCl} + m_{AgNO_3}$$

$$m_{sol} = 2$$

[Molarity $\approx$ Molality]

$$\Delta T_f = K_f \cdot m \cdot i$$

$$\Delta T_f = 1.86 \cdot 2 \cdot \frac{1}{2} \text{ K}$$

$$\Delta T_f = 1.86 \text{ K}$$

Ans. A.

4.

Boiling point: When vapour pressure becomes equal to atmospheric pressure.

A. Incorrect. As the vapour pressure of P becomes equal to 1atm earlier than Q and R.

Correct order:

$$V.P-P > Q > R$$

$$B.P-P < Q < R$$

B. Incorrect. $\frac{dP}{dT} = \text{slope}$

$$\text{At } T=0, \frac{dP}{dT} = \frac{K}{(\text{tends to } 0)^2} = \infty$$

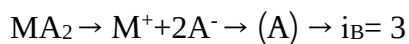
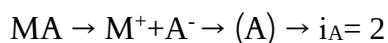
But in the graph, the slope at $T=0$ is zero.

C. Incorrect. Intermolecular interaction : $P < Q < R$

D. Correct. The boiling point of a liquid is the temperature at which its vapour pressure equals the external pressure. In this graph, we can observe the vapour pressures of the three liquids at different temperatures. At 101.3 kPa (1 atm), the boiling points can be determined from the graph. From the graph, the vapour pressure of liquid Q reaches 101.3 kPa at a temperature close to 65°C, indicating that the normal boiling point of liquid Q is close to 65°C.

Ans. D.

5.



$$\Delta T_b = i \cdot k_b \cdot m$$

i = Van't Hoff factor

k_b = molal elevation constant, m = molality

$$\frac{\Delta T_b(A)}{\Delta T_b(B)} = \frac{i_A}{i_B} \cdot \frac{m_A}{m_B} \cdot \frac{k_b}{k_b}$$

$$m_A = \frac{\text{moles}}{\text{in 1 kg solvent}} = \frac{W_2/X \text{ moles}}{W_1 \cdot 10^{-3} \text{ kg}}$$

$$m_B = \frac{W_2/2X \text{ moles}}{W_1 \cdot 10^{-3} \text{ kg}}$$

$$\frac{m_A}{m_B} = \frac{1}{1/2} = 2$$

$$\frac{i_A}{i_B} = \frac{2}{3}$$

$$\text{Therefore, } \frac{\Delta T_b(A)}{\Delta T_b(B)} = \frac{2}{3} * 2 = \frac{4}{3}$$

Ans. B

6.

The elevation in boiling point is given by:

$$\Delta T_b = i K_b m$$

For dissociation, the degree of dissociation (α) is given by:

$$\alpha = (i - 1) / (n - 1)$$

For $n = 2$, we have:

$$\text{So } \alpha = i - 1 \Rightarrow i = 1 + \alpha$$

The depression in freezing point is given by:

$$\Delta T_f = i K_f m$$

For association, α is given by:

$$\alpha = (1 - i) / (1 - 1/n)$$

The total van't Hoff factor (i) becomes:

$$i = 1 - \alpha(1 - 1/n)$$

Now, taking the ratio of boiling point elevation and freezing point depression, we get:

Simplifying, we have:

$$\frac{\Delta T_b}{\Delta T_f} = \frac{i K_b m}{i K_f m}$$

Multiplying K_f on both sides, we get:

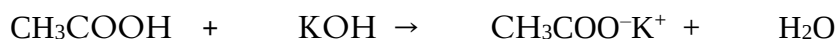
$$\frac{\Delta T_b K_f}{\Delta T_f} = \frac{(1 + \alpha) K_b}{1 - \alpha \left(1 - \frac{1}{n}\right)}$$

On rearranging, we get the final expression for n :

$$n = \frac{\Delta T_b K_f}{\Delta T_f (1 + \alpha) K_b}$$

Ans. C.

7.



1 mole 1 mole

$$\begin{aligned}\text{No. of mole} &= \frac{m.V(\text{ml})}{1000} \\ &= \frac{0.5 \times 30}{1000} \\ &= 15 \times 10^{-3} \text{ mole}\end{aligned}$$

1 mole KOH $\rightarrow$ 1 mole CH₃COOH = 15×10^{-3} mole

15×10^{-3} mol KOH $\rightarrow$ 15×10^{-3} mole CH₃COOH used

Mass of CH₃COOH = 60

$$\begin{aligned}\text{Mass of CH}_3\text{COOH} &= 15 \times 10^{-3} \times 60 \\ &= 900 \times 10^{-3} \\ &= 0.9 \text{ g of CH}_3\text{COOH}\end{aligned}$$

$$\begin{aligned}\% \text{ purity of CH}_3\text{COOH} &= \frac{w_{\text{reacted}}}{w_{\text{given}}} \times 100 \\ &= \frac{0.9}{15} \times 100 \\ &= 60\%\end{aligned}$$

Ans. C.

8.

From the diagram and given conditions:

- The three curves X, Y, and Z represent the vapour pressures of the three liquids as functions of temperature.
- P₁, P₂, and P₃ are different pressures, with P₁ being 1 atm.
- T₁, T₂, and T₃ are different temperatures.

Key Points:

1. Pure water has the highest vapour pressure among the three liquids because it has no solute particles to lower its vapour pressure.
2. The vapour pressure of the dilute NaCl solution is the lowest because NaCl dissociates into more particles (ions), leading to a greater reduction in vapour pressure due to the colligative properties of solutions.

3. The glucose solution has a vapour pressure between pure water and the NaCl solution, as glucose does not dissociate into ions.

In the diagram:

- Curve X (highest vapour pressure) should correspond to pure water.
- Curve Y (intermediate vapour pressure) should correspond to the glucose solution.
- Curve Z (lowest vapour pressure) should correspond to the NaCl solution.

With these points in mind, let's analyze the statements:

1. At ($T = T_1$), the vapour pressure (P_1) is for water, and it is the boiling point of water at 1 atm pressure.
2. The vapour pressures of the glucose and NaCl solutions at ($T = T_1$) will be lower than (P_1).
3. Similarly, at ($T = T_2$) and ($T = T_3$), the relationships between the vapour pressures of the three solutions will remain the same.

Based on the above analysis, the correct interpretation of the diagram is:

- Curve X corresponds to pure water.
- Curve Y corresponds to the glucose solution.
- Curve Z corresponds to the NaCl solution.

Given these interpretations, the correct statements are:

1. Curve X corresponds to pure water.
2. Curve Y corresponds to the glucose solution.
3. Curve Z corresponds to the NaCl solution.

These statements are consistent with the colligative properties of solutions and the observed vapour pressure trends.

Ans. A, B, C

To find the elevation in boiling point ΔT_b in terms of the boiling point elevation constant K_b , we need to use the formula for boiling point elevation in solutions, which is:

$$\Delta T_b = K_b \cdot m \cdot i$$

ΔT_b is the elevation in boiling point.

K_b is the boiling point elevation constant.

m is the molality of the solution.

i is the van't Hoff factor, which represents the number of particles into which a solute dissociates in solution.

Given the details of the problem:

- 0.1 mol of weak acid HX with a degree of ionization of 0.1.
- 0.2 mol of weak acid HY with a degree of ionization of 0.2.
- Dissolved in 1 kg of water.

1. Calculate the effective number of particles (i for each solute):**

For HX:

$$i_{HX} = 1 + \alpha_{HX} = 1 + 0.1 = 1.1$$

Where α_{HX} is the degree of ionization for HX.

For HY:

$$i_{HY} = 1 + \alpha_{HY} = 1 + 0.2 = 1.2$$

Where α_{HY} is the degree of ionization for HY.

Calculate the molality m of each solute:

Since the solutes are dissolved in 1 kg of water, the molality is equal to the number of moles per kilogram of solvent:

For HX:

$$m_{HX} = 0.1 \text{ mol/kg}$$

For HY:

$$m_{HY} = 0.2 \text{ mol/kg}$$

Calculate the total change in boiling point:

Using the boiling point elevation formula for each solute, we sum the contributions from each solute:

For HX:

$$\Delta T_{bHX} = K_b \cdot m_{HX} \cdot i_{HX} = K_b \cdot 0.1 \cdot 1.1 = 0.11K_b$$

For HY:

$$\Delta T_{bHY} = K_b \cdot m_{HY} \cdot i_{HY} = K_b \cdot 0.2 \cdot 1.2 = 0.24K_b$$

Total boiling point elevation:

$$\Delta T_b = \Delta T_{bHX} + \Delta T_{bHY} = 0.11K_b + 0.24K_b = 0.35K_b$$

Thus, the correct option is:

$$\Delta T_b = 0.35K_b$$

Ans. B.